Medical Microbiology

Lecture Two

Microbial growth(Neutralization)

Third stage- College of Dentistry

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Growth of Microbes:

- One of the most critical factors for microbial growth is the availability of nutrients and energy.
- Microbes grow via binary fission, resulting in exponential increases in numbers of cells.
- Bacterial growth means an increase in the number of cells, not an increase in cell size.
- One cell becomes colony of millions of cells.
- These Bacterial cell produce identical offspring, which cannot be distinguished as a parents or offspring.

Generation time:

- **Is the time it takes for a single cell to grow and divide.**
- **Average for most bacteria is 1-3 hours.**
- ***Escherichia coli*** **: 20 minutes → 20 generations about (7 hours), 1 cell becomes 1 million cells.**
- ***Mycobacterium*** **much slower generation time: about (12-24) hours.**

Phases of Growth:

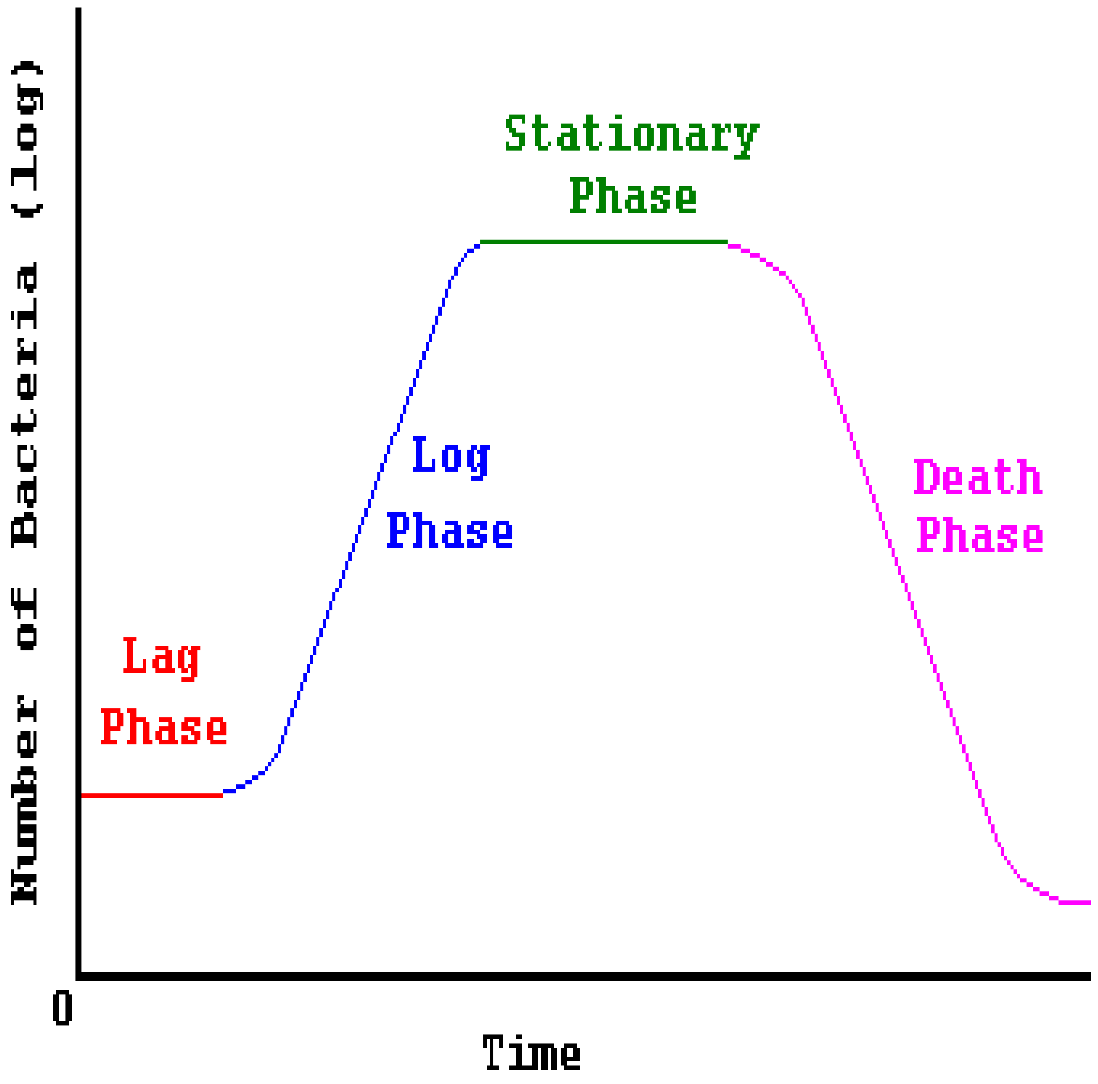
Four main growth phases:

1-Lag phase.

2-Exponential (Log) phase.

3-Stationary phase.

4-Decline phase (death phase).



1-lag phase:

- Bacteria adapt themselves to growth conditions.
- This period where the individual bacteria are maturing and not yet able to divide.
- During this phase cells are not dormant and change very little because the cells do not immediately reproduce in a new medium.
- During the lag phase synthesis of RNA, enzymes and other molecules occurs.

2-The log phase:

- (sometimes called the logarithmic phase or the exponential phase): is a period characterized by cells doubling.
- The number of new bacteria appearing per unit time

  is proportional to the present population.
- If no limiting factors are present, doubling will continue at a constant rate so both the number of cells and the rate of population increase doubles with each consecutive time period.
- The Exponential growth phase cannot continue indefinitely, because the medium is soon depleted of nutrients and enriched with wastes.

3-The stationary phase:

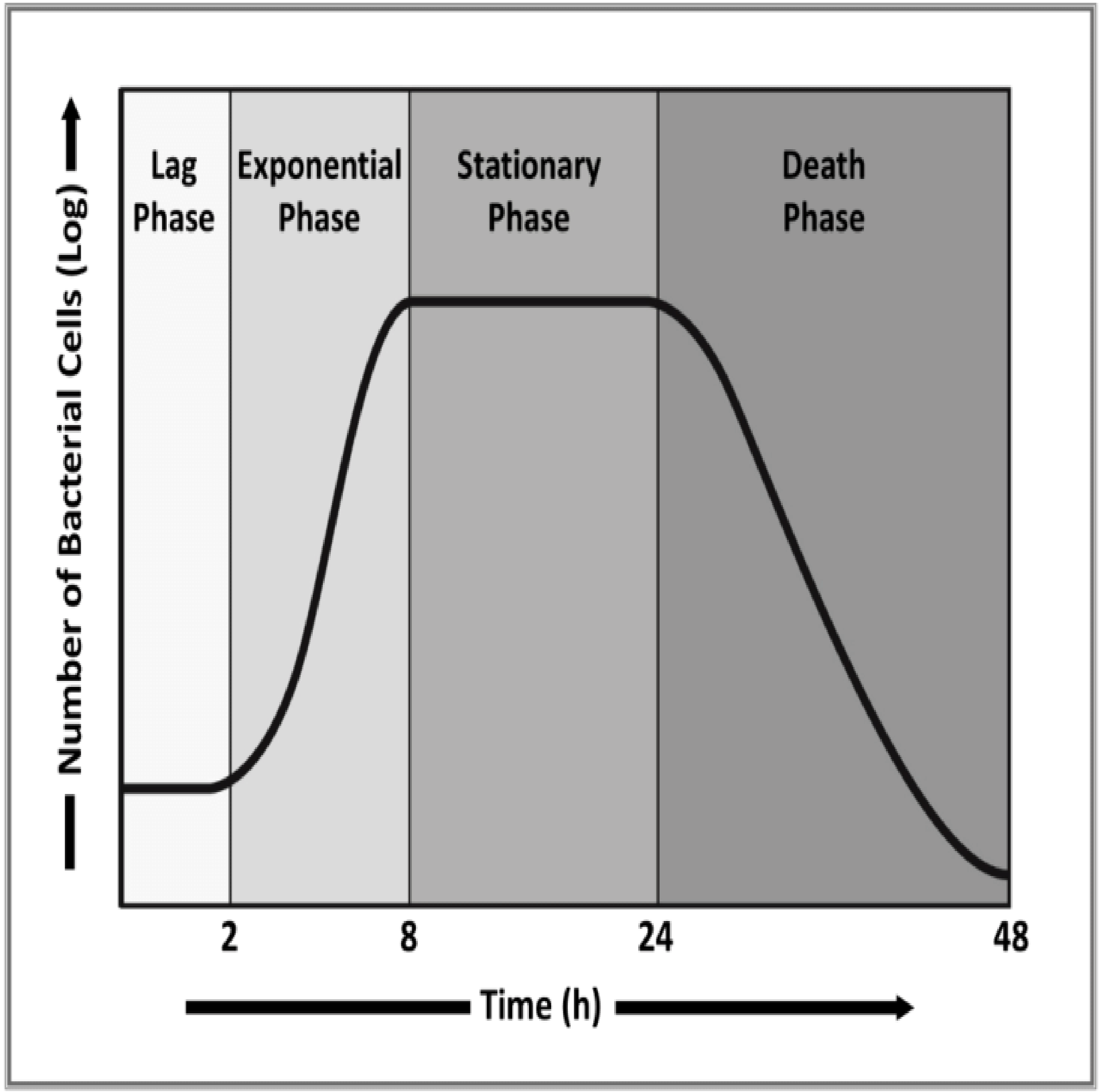
- Is often due to a growth limiting factor such as the depletion of an essential nutrient, and the formation of an inhibitory product such as an organic acid.
- Stationary phase results from a situation in which growth rate and death rate are equal.
- Mutations can occur during stationary phase.
- DNA damage is responsible for many of the mutations

arising in the genomes of stationary phase or starving bacteria.

4-death phase (decline phase):

Bacteria die, this could be caused by lack of nutrients,

Environmental temperature above or below the tolerance band for the species, or other injurious conditions.



Requirements for Growth:

- Bacteria must obtain or synthesize Amino acids, carbon sources, nitrogen sources, moister, inorganic salts and trace elements

Carbohydrates & Lipids to build up the cell.

Key Requirement of growth included:

1. Nutrients.
2. Growth factors.
3. Temperature.
4. Oxygen.
5. PH (potential of hydrogen).
6. Osmotic pressure.

-Growth requirements & metabolic secondary products enable to Classify different types of bacteria.

Nutritional types of bacteria:

A. Depend on how the organism obtains carbon for synthesizing cell mass divided into:

1- Autotrophic – carbon is obtained from carbon dioxide (CO_2) from atmosphere.

2- Heterotrophic – carbon is obtained from organic compounds.

3- Auto-heterotrophic – carbon is obtained from both organic compounds and CO_2.

B. Depend on how the organism obtains reducing equivalents used either in energy conservation or in biosynthetic reactions:

1- lithotrophic – reducing equivalents are obtained from inorganic compounds.

2- Organotrophic – reducing equivalents are obtained from organic compounds.

C. Depend on how the organism obtains energy for living and growing:

1- Chemotrophic – energy is obtained from chemical Compounds.

2- Phototrophic – energy is obtained from light.

- Dependent on the basis of this can be divided into many sources:

1-Chemolithoautotrophs: obtain energy from chemical compounds, reducing equivalents from inorganic compounds and carbon from CO_2 . e.g.: Knallgas-bacteria (hydrogen-oxidizing bacteria)

2- Photolithoautotrophs: obtain energy from light, reducing equivalents from inorganic compounds and carbon from CO_2. e.g.: Cyanobacteria

3- Chemolithoheterotrophs: obtain energy, chemical compounds and reducing equivalents from inorganic compounds, carbon by organic compounds . e.g.: *Nitrobacter* spp

4- Chemoorganoheterotrophs: obtain energy, carbon, and reducing equivalents from organic compounds. e. g. *Escherichia coli*

2-Growth Factors:

- Bacteria may require small amount of certain organic compound for growth.
- These compounds are essential substances that the organism is unable to synthesize from available nutrient because they lack the genetic and metabolic mechanisms to synthesize them.

Growth factors are organized into three categories:

1-Purines and pyrimidines: required for synthesis of DNA and RNA.

2-Amino acids: required for the synthesis of proteins.

3-Vitamins: needed as coenzymes and functional groups of certain enzymes.

3.Temperature:

1- **Psychrophiles:** cold-loving, can grow at 0 C. e.g. Flavobacterium.

2- **Mesophiles:** moderate temperature-loving (Most bacteria)

Include most pathogens. e.g. Staphylococcus aureus.

- Best growth between 25 to 40 C.
- Optimum temperature commonly 37C.
- Many have adapted to live in the bodies of human.

3- **Thermophiles:** heat-loving e.g. Thiothrix calidifontis.

Optimum growth between 50 to 80 C.

- Many cannot grow below 45 C.

-Adapted to live in sunlit soil and hot springs.

4.Oxygen:

(a) Obligate aerobes – require O2. e.g. *Bacillus subtilis.*

(b) Obligate anaerobes – die in the presence of O2. e.g. *Clostridium tetani*.

(c) Facultative anaerobes – can use O2 but also grow without it. e.g. *Yersinia pestis*.

(d) Microaerophilic -requires lower oxygen to survive. e.g. *Helicobacter pylori*.

(e) Aerotolerant anaerobe: tolerate the presence of oxygen but does not require it for its growth. e.g. *Streptococcus pyogenes*.

5. PH:

Organisms can be classified as:

A-Acidophilus: “Acid loving”.

- Grow at very low pH (0.1 to 5.4) (many fungi).

B-Neutrophils':

- Grow at pH (5.4 to 8.5) includes most human pathogens.

C-Alkaliphiles: “Alkali loving”.

- Grow at alkaline or high pH (7 to 12 or higher)
- *Vibrio cholera* -optimal pH 9.
- Soil bacterium Agrobacterium grows at pH 12.

6. Osmotic Pressure:

- Microbes require minerals or nutrients for their growth, which can be obtained from the surrounding solutions.
- Osmotic pressure and salt concentration of the solution can influence bacterial growth.
- The bacterial cell wall gives a mechanical strength that allows the bacteria to withstand alternations in the osmotic pressure.

Media for bacterial growth:

For identification of bacteria, the organisms are obtained by growing on artificial culture media.

Types of culture media:

- **Simple media.**
- **Enriched media.**
- **Selective media.**
- **Differential media.**

1. Simple media:

- Contain basic nutrients for bacterial growth like broth with peptone, e. g. Nutrient broth.

2. Enriched media:

- Enriched by some substances like: Blood & Serum,

e.g. Blood agar, Chocolate agar.

3. Selective media:

- Contain substances such as bile salts or antibiotics

that inhibit the growth of some organisms but have little or no effect on the required organism, e. g. Salmonella Shigella agar.

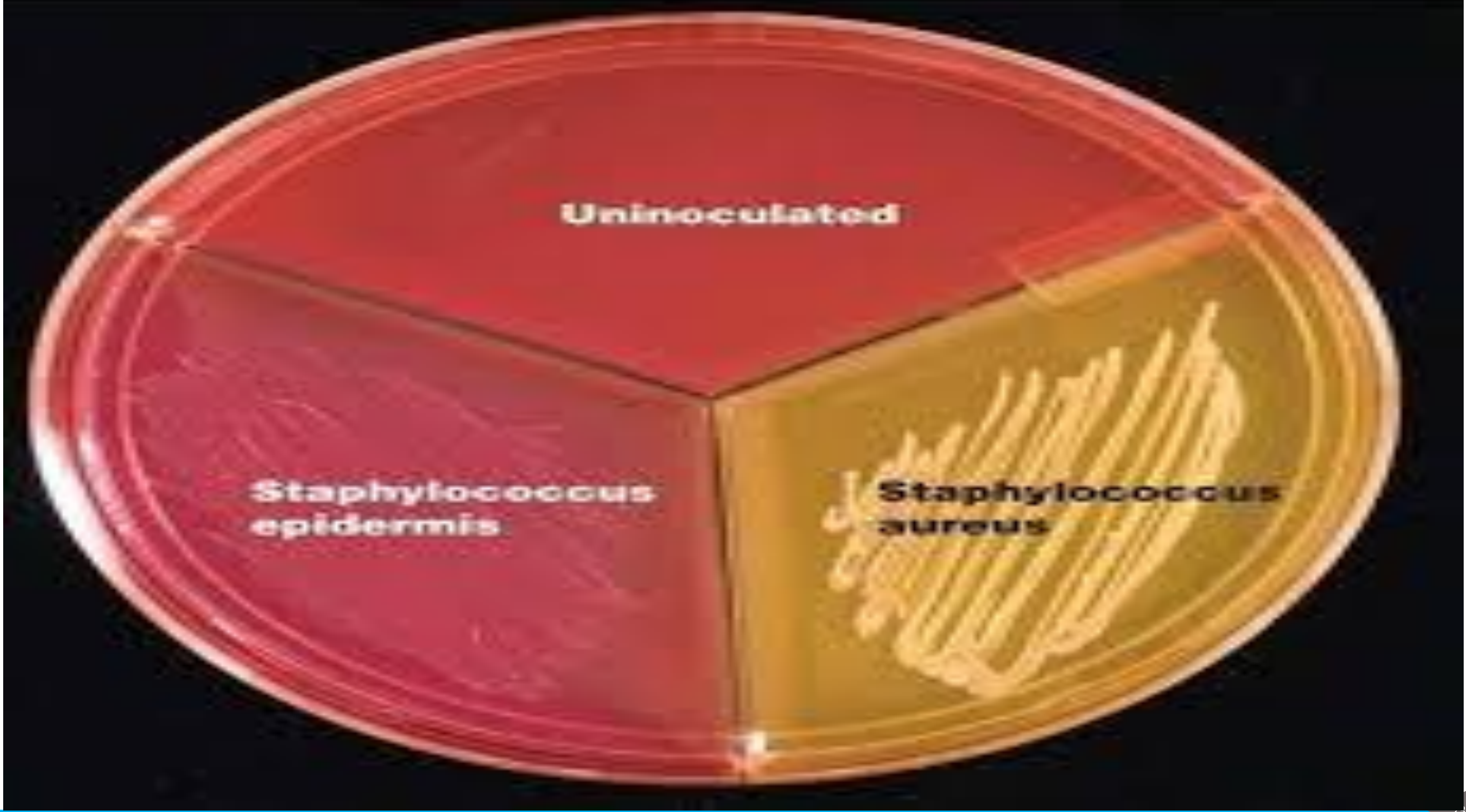
4. Differential media

- **Contains specific ingredients or chemicals that allow the observer to visually distinguish which species possess and which species lack a specific biochemical process.**

e.g. MacConkey agar.

- **Mannitol salt agar or MSA is a commonly used selective and differential growth medium in microbiology, the high salt content of 7.5% inhibits most bacteria other than staphylococci. e.g. *Staph. aureus***


SS
Agar


Uninoculated
Staphylococcus epidermis
Staphylococcus aureus

Usage of bacterial culture:

1-Diagnosis, prevention and treatment of infection Diseases.

2-Characterization of bacteria.

3-Preparation of vaccines, toxoids and other biologic Products.

4-Application in industry and agriculture.

5-Uses for genetic engineering.

Any Question

Thank you